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2.1.2. Diploblastic and Triploblastic organisation

During embryonic development, the tissues and organs of animals originate from two or three embryonic germ layers. On the basis of the origin and development, animals are classified into two categories: Diploblastic and Triploblastic.

Animals in which the cells are arranged in two embryonic layers (Figure 2.1), the external ectoderm, and internal endoderm are called **diploblastic animals**. In these animals the ectoderm gives rise to the epidermis (the outer layer of the body wall) and endoderm gives rise to gastrodermis (tissue lining the gut cavity). An undifferentiated layer present between the ectoderm and endoderm is the mesoglea. (Corals, Jellyfish, Sea anemone)

Animals in which the developing embryo has three germinal layers are called **triploblastic** animals and consists of outer ectoderm (skin, hair, neuron, nail, teeth, etc), inner endoderm (gut, lung, liver) and middle mesoderm (muscle, bone, heart). Most of the triploblastic animals show organ system level of organisation (Flat worms to Chordates).

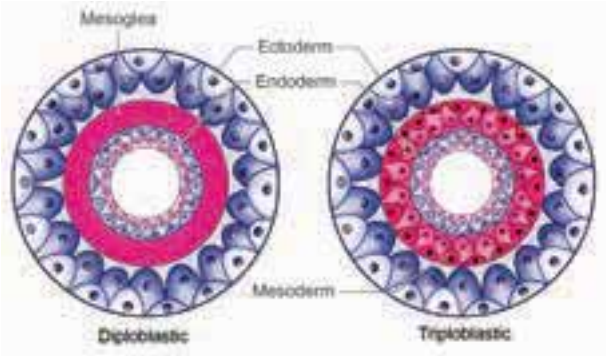

Figure 2.1 Germinal layers

2.1.3. Patterns of symmetry

Symmetry is the body arrangement in which parts that lie on opposite side of an axis are identical. An animal's body plan results from the animal's pattern of development. The simplest body plan is seen in sponges (Figure 2.2). They do not display symmetry and are **asymmetryical**. Such animals lack a definite body plan or are irregular shaped and any plane passing through the centre of the body does not divide them into two equal halves (Sponges). An asymmetrical body plan is also seen in adult gastropods (snails).

Figure 2.2. Asymmetry in sponges

Symmetrical animals have paired body parts that are arranged on either side of a plane passing through the central axis. When any plane passing through the central axis of the body divides an organism into two identical parts, it is called **radial symmetry**. Such radially symmetrical animals have a top and bottom side but no dorsal (back) and ventral (abdomen) side, no right and left side. They have a body plan in which the body parts are organised in a circle around an axis. It is the principal symmetry in diploblastic animals. Cnidarians such as sea anemone and corals (Figure 2.3) are radially symmetrical. However, triploblastic animals like echinoderms (e.g., starfish) have five planes of symmetry and show **Pentamerous radial symmetry**.